

WEATHER TREND FORECASTING

**A Project Completed Under the PM Accelerator
Program**

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PM Accelerator Mission

Empowering professionals with the skills, mentorship, experience, and network needed to build successful careers in product management and technology.

Project Overview

Objective:

This project focuses on building a machine learning model to predict weather-related outcomes using historical weather data

Dataset used:

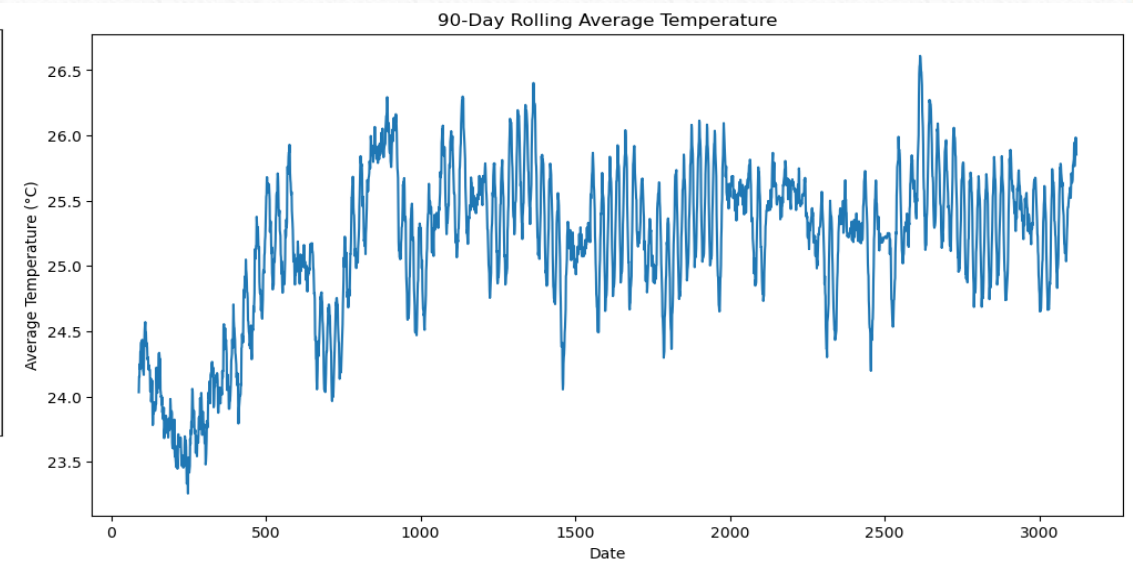
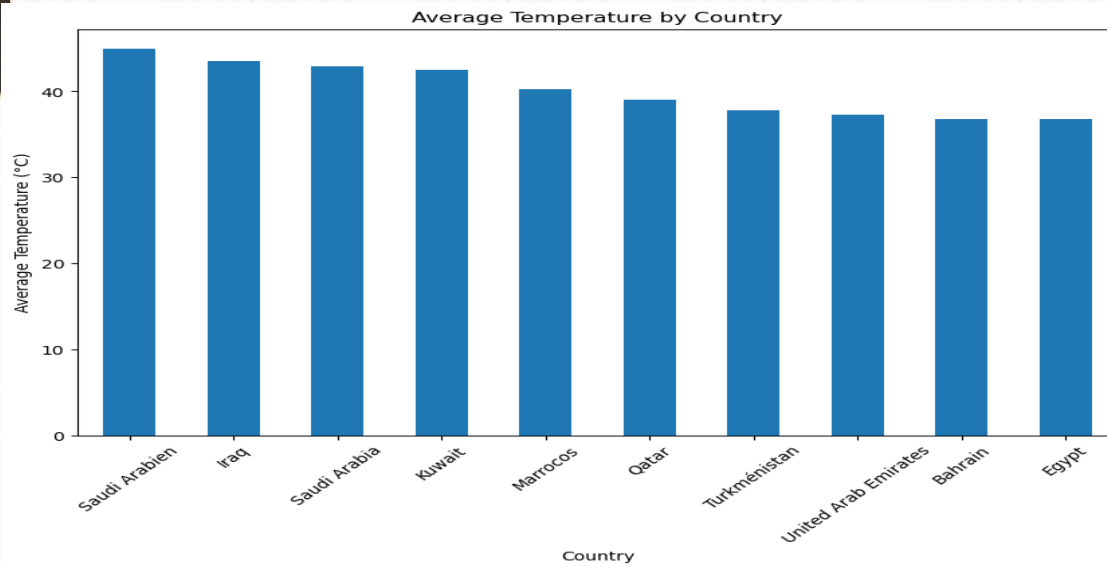
Global weather repository from Kaggle; contains features like temperature, atmospheric pressure, country, precipitation etc.

Why it matters:

- Improves disaster preparedness
- Supports climate planning

EDA(Exploratory Data Analysis)

Advanced EDA, which included time-based analysis, seasonal pattern identification, rolling statistics and geographical comparison

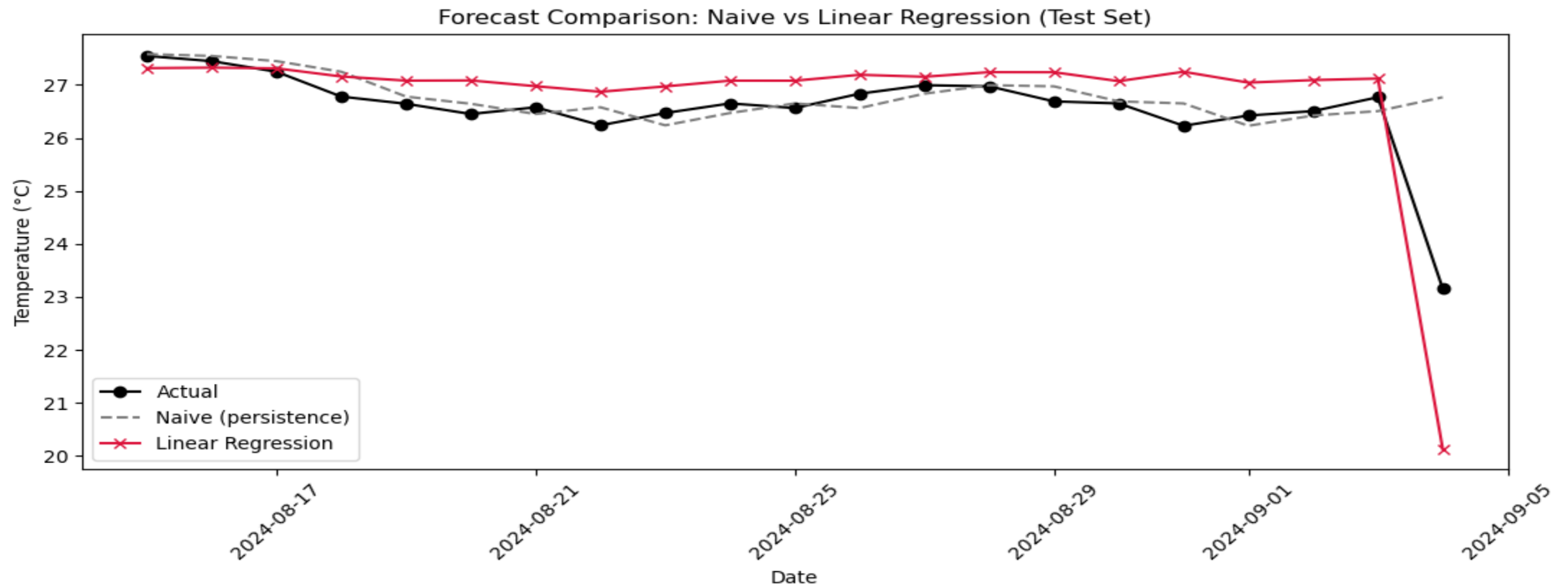


Modeling

Models used:

- Linear regression
- Naïve Persistence baseline
- Random Forest
- Arima
- Xgboost – Best performing, R-squared of 0.878

Model Performance



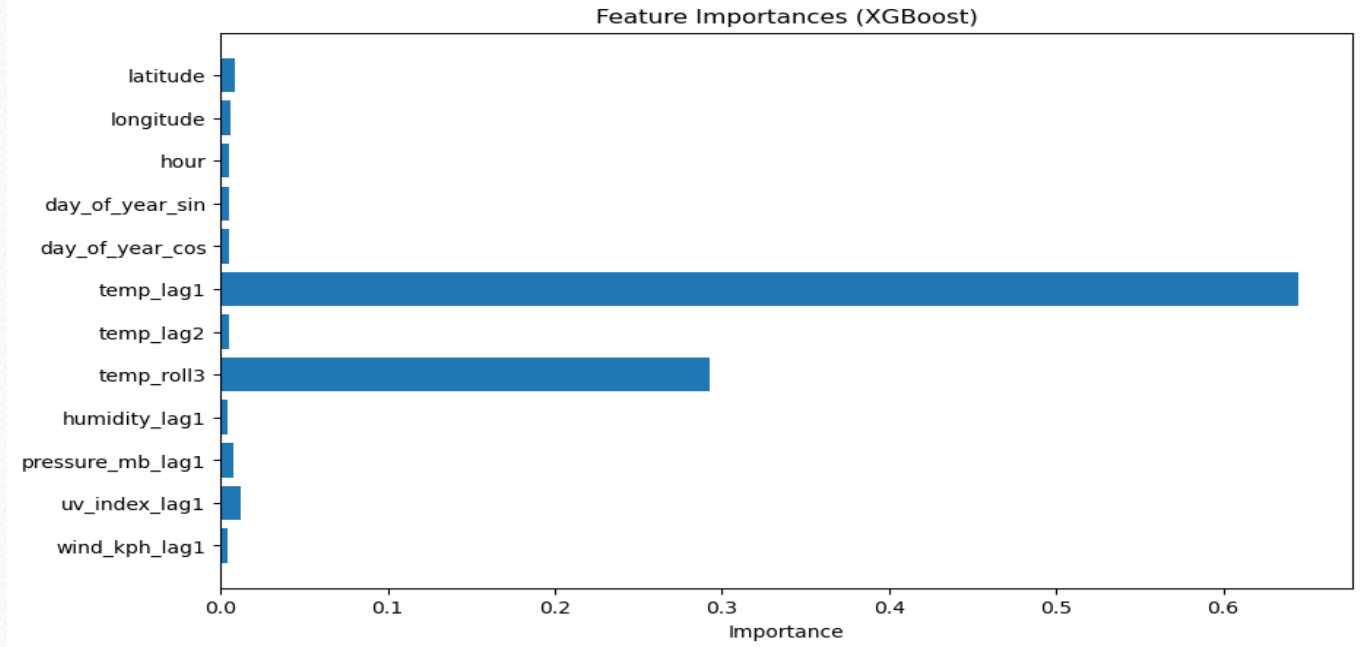
Interpretation

The above graph is a comparison between Naïve and linear regression compared to the actual values .The best performing model is XGboost. The model achieved an R-squared of 0.878 meaning its very effective at weather forecasting.

Feature Importance; XGboost

Identified 3 main features that affected the weather. They include;

- Temp_lag1
- Temp_roll3
- Uv_index_lag1



Recommendations

- Collect more historical data to improve model performance
- Continue monitoring model accuracy as new data becomes available
- Deploy the model to support future decision making

THANK YOU